

GARMIN®

AVIATION TRAINING

GFC 500

Working with the Addendum

For Reference Only
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Garmin Aviation Training is intended to provide the best practices for the installation, use and maintenance of Garmin Aviation products. It is incumbent on technicians and pilots to ensure that the instructions contained in the most recent versions of Garmin product manuals, bulletins and notifications are followed.

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Objectives

Monday, March 30, 2026

- ✈ Update on SLEP: Tick-tock
- ✈ Documentation:
 - ✈ Pre-Install Checklist
 - ✈ Master Drawing List
 - ✈ Model Specific Addendums
 - ✈ Cessna 210-5 (Most Recent)
 - ✈ Cessna 172 E/S (First Issuance)
 - ✈ Gains Addendum
 - ✈ Aircraft Flight Manual Supplement

Objectives

Monday, March 30, 2026

- ✈ After completing this section, you will:
 - ✈ Understand the breakdown of Garmin's GFC 500 Model Specific Addendums for Autopilot Installation.
 - ✈ Better articulate the common faults as presented by the notes and advice by Garmin Autopilot Subject Matter Experts.
 - ✈ Identify work processes for better installation experiences.

- ✈ Certified SB 23024 & Experimental SB 23038
- ✈ To improve GSA reliability.
- ✈ Several design and manufacturing process improvements led to the current version of GSA 28 (-20/-21 Mod 1), which began shipping in September 2022.
- ✈ Service Life Extension Program no cost exchanges for Mod 1 are allowed for 5 years (4May23 – 4May28).

- ✈ NOH replacement servos will have the greater of either:
 - ✈ the remaining 2yr factory warranty or
 - ✈ standard 6-month repair/exchange warranty.
- ✈ It is important to promptly return 'removed servos', as these will be upgraded to Mod 1 units and eventually sent back out.
- ✈ RMA submissions should have the planned install date for when they need servos.

- ✈ Updated RMA (Return Material Authorization) form MUST be used.
 - ✈ No longer need 'GSA28 RMA Scheduling Form' as the initial rush for program adoption is past.
- ✈ Like other SB, warranty labor will be paid to dealers, but not to experimental customers.
- ✈ Experimental RMA: G3Xpert@garmin.com



- ✈ RMA Forms: avionicsrma.americas@garmin.com
- ✈ Warranty Claims or Questions: warranty@garmin.com
- ✈ For inquiries, scheduling, and customers who have already paid for out of warranty replacement:
GSAinquiries@Garmin.com
- ✈ Experimental RMA: G3Xpert@garmin.com
- ✈ Customer Support Articles can be found on the public facing website: www.Garmin.com/GSA28Info.

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NOTES

Ends May 4, 2028

All affected servo units are covered under by the program for a period of five (5) years from the date of the service bulletin (May 4 2023).

Labor reimbursement for installations in certified aircraft performed by an Authorized Garmin Dealer ONLY (2.5 hrs per Servo).

Garmin Aviation Dealer Academy | Garmin GSA 28 Service Life Extension Course

garmin.sharepoint.com/sites/AviationTrainingVideos/SitePages/GSA-28---US.aspx

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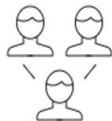
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✈️ Pre-Install Checklist

- ✈️ Aircraft Photos?
- ✈️ Other Modifications?
- ✈️ Garmin Documents
 - ✈️ Approved Models List
 - ✈️ Price Catalog
 - ✈️ Master Data List
 - ✈️ Addendums
 - ✈️ Flight Manual Supplements
 - ✈️ Installation Documentation

✈️ Post-Install Checkout

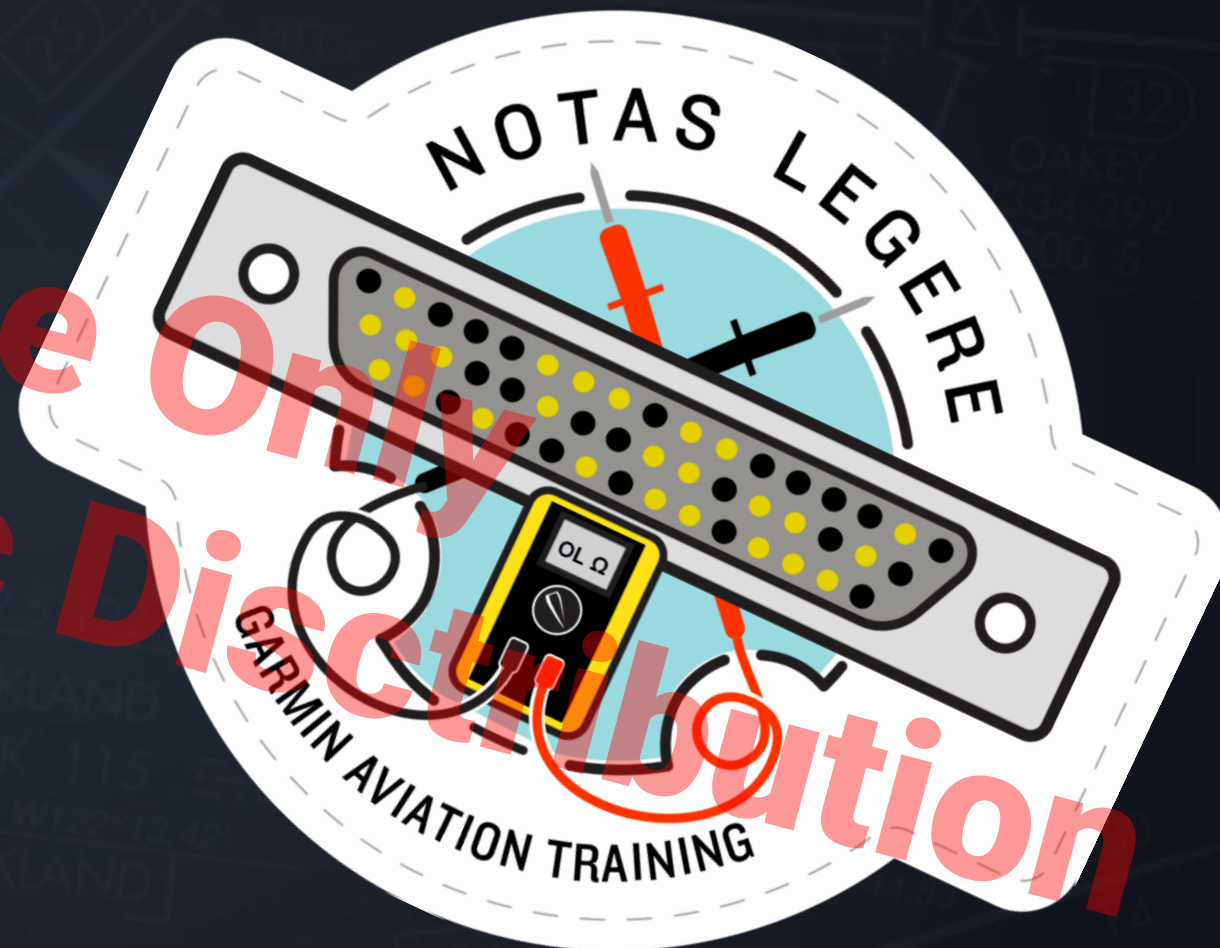


✈️ Where to start?

- ✈️ Approved Model List and Autopilot Price Catalog
- ✈️ Price Catalog
- ✈️ Master Drawing List
- ✈️ Type-specific Addendums and AFMS



- ✈ Double check your MDL
 - ✈ Ensure you're using the most recent documentation for your airframe!
- ✈ For every 10 minutes you spend reading the notes on the Model Specific Addendum You'll save an HOUR of troubleshooting your installation!
 - ✈ We never have time to do it 'right', but we always make time to do it again!



Master Drawing List, GFC 500, Part 23 AML STC	005-01264-00	87	February 1, 2024	Download
Master Drawing List, GFC 500, Part 23 AML STC	005-01264-00	88	March 1, 2024	Download
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Maintenance Manual w/ICA - STC Part 23, GFC 500	190-02291-01	1	November 1, 2017	Download

✈️ Current MDL is revision 99

✈️ Always check for updates before delivery of the aircraft!

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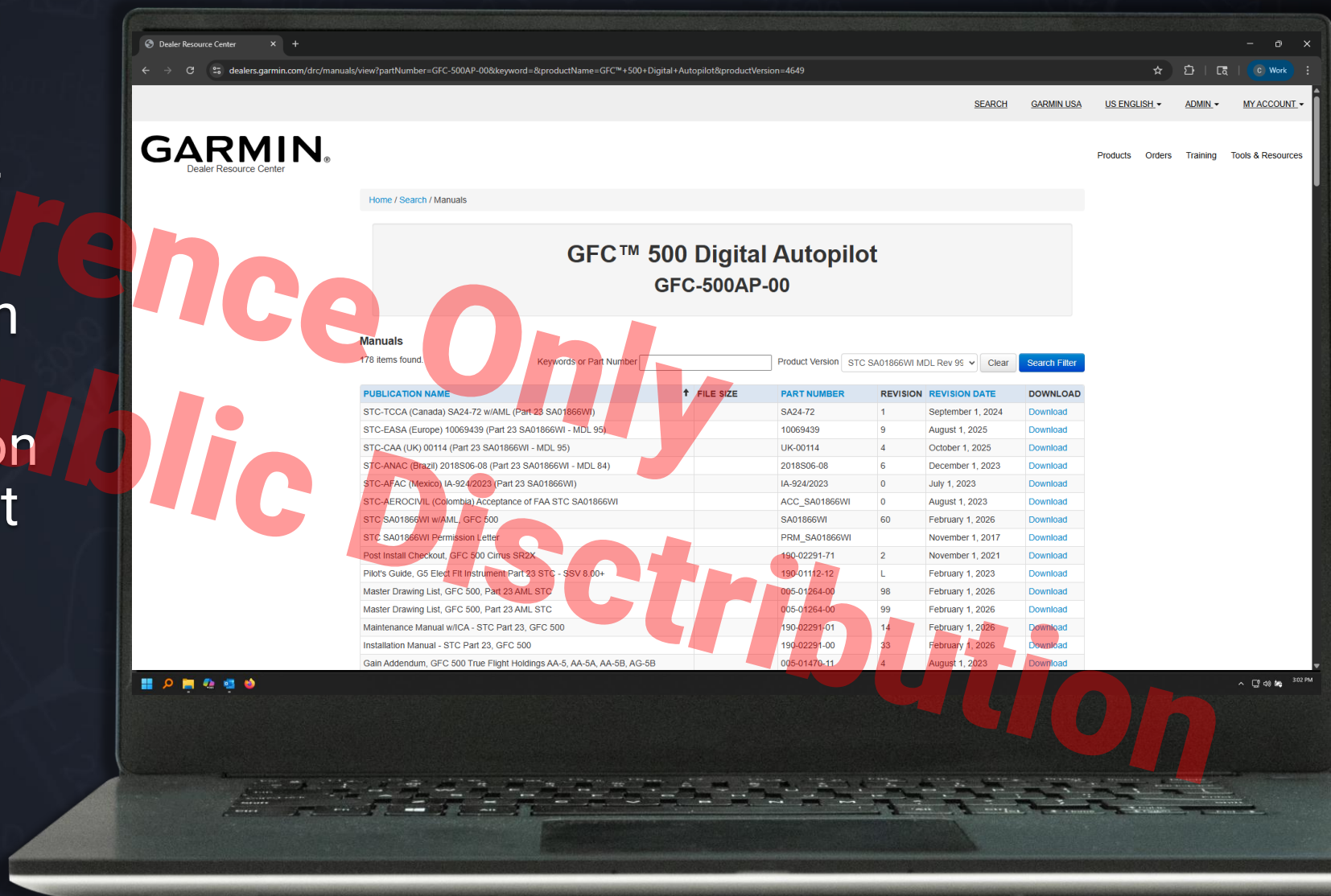
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✈️ Using the product version filter

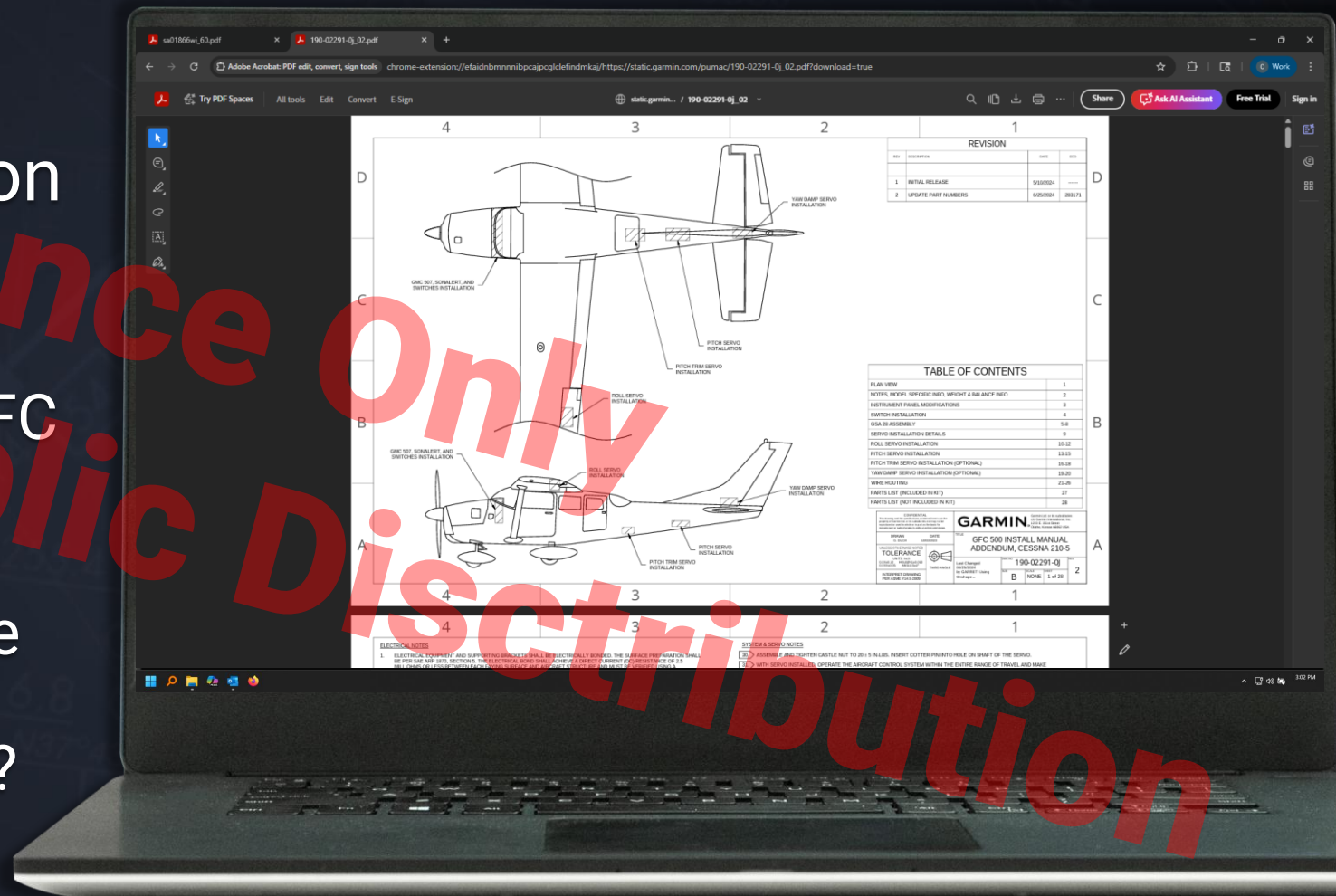
- ✈️ Offers simplicity for quick reference
- ✈️ Works for all Garmin STC products
- ✈️ Focuses searches on only the most recent documents



✈️ Notes offer a direct insight into the lessons learned by the certification team during initial issuance of the STC

✈️ Newest addition to the GFC 500 AML is the Cessna 210-5(A)

✈️ How many notes does the Cessna 210-5 (A) Model Specific Addendum offer?

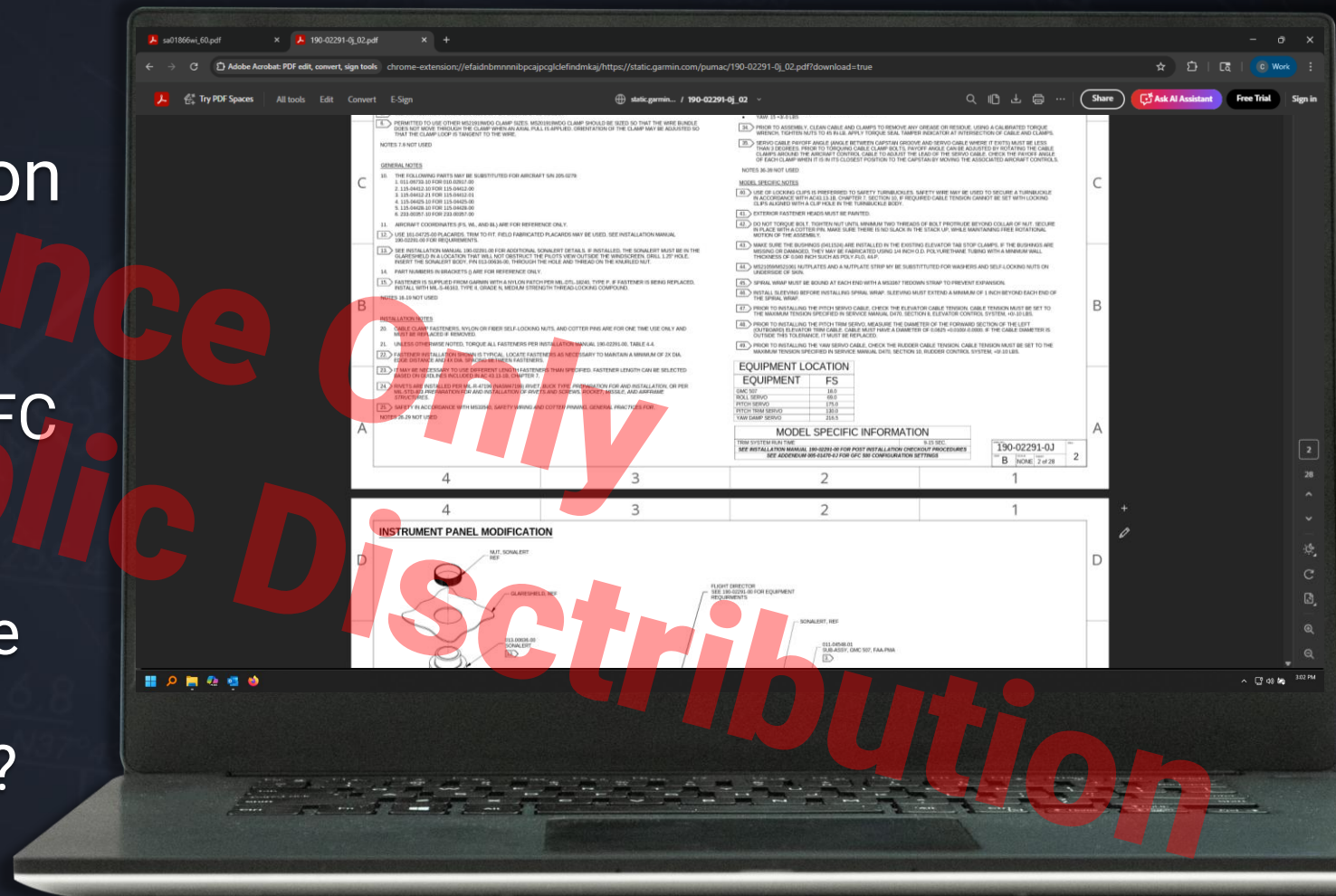


✈️ Notes offer a direct insight into the lessons learned by the certification team during initial issuance of the STC

✈️ Newest addition to the GFC 500 AML is the Cessna 210-5(A)

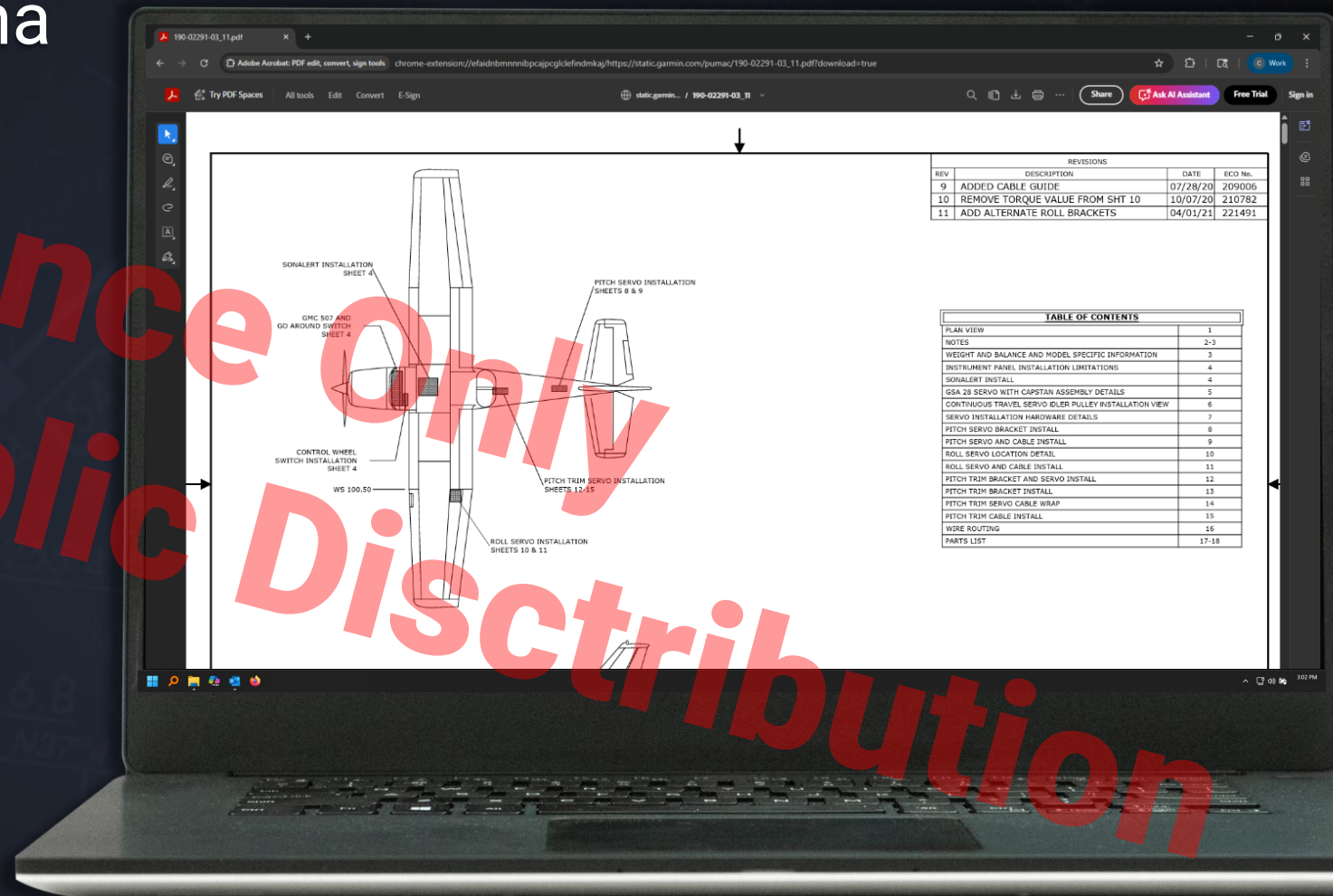
✈️ How many notes does the Cessna 210-5 (A) Model Specific Addendum offer?

✈️ 49 as of Rev 2



✈️ Let's compare the Cessna 172 E/S Model initial issuance from 2017?

✈️ How many notes?



✈️ Let's compare the Cessna 172 E/S Model initial issuance from 2017?

✈️ How many notes?

✈️ 31 as of Rev 11

190-02291-03_11.pdf

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NOTES (CONTINUED)

20. TEMPORARILY CLAMP SERVO AND BRACKET ASSEMBLY IN PLACE AND ENSURE EACH SERVO CABLE PAYOFF ANGLE (ANGLE BETWEEN THE CABLE AND CAPSTAN GROOVE WHERE IT EXITS) IS LESS THAN 3 DEGREES. MINOR BRACKET POSITIONING ADJUSTMENTS ARE ACCEPTABLE TO ACHIEVE THE REQUIRED SERVO CABLE PAYOFF ANGLE.

21. PART OF 011-02951-10 SUB-ASSY, CAPSTAN KIT, 1/16" SPIRAL, 5 GROOVE, GSA 28.

22. PART OF 011-02952-04 SUB-ASSY, GSA 28 MOUNTING KIT, CONTINUOUS TRAVEL.

23. INSTALL PITCH TRIM CABLE EXTENSION AFTER POSITIONING PITCH TRIM SERVO BRACKETS.

24. SEPARATION OF PITCH TRIM CABLES WILL BE LESS THAN 1/8 INCH. ADJUST ROTATION OF TRIM STOP BLOCKS TO MAXIMIZE SEPARATION OF PITCH TRIM CABLES THROUGHOUT THE ENTIRE LENGTH OF TRAVEL.

25. AFTER PITCH TRIM SERVO INSTALLATION IS COMPLETE, RUN PITCH TRIM FROM STOP TO STOP A MINIMUM OF FIVE TIMES. AFTER RUNNING PITCH TRIM FROM STOP TO STOP, VERIFY PITCH TRIM CABLES ARE MAINTAINING MAXIMUM POSSIBLE SEPARATION FROM EACH OTHER AFT OF THE PITCH TRIM SERVO. ADJUST ROTATION OF TRIM STOP BLOCKS TO MAXIMIZE SEPARATION OF PITCH TRIM CABLES IF NEEDED.

26. PART OF 011-04755-00 SUB-ASSY, CONN KIT, GSA 28.

27. FABRICATED SHIM BETWEEN 115-02737-00 AND 115-02738-00: 6061-T6 OR 2024-T3, 0.040" THICK, CLEAR CHEMICAL CONVERSION COATING, MIL-DTL-5541 TYPE II, CLASS 3. SHIM DIMENSIONS ARE APPROXIMATELY 1.25" X 1.25" AND IS APPROXIMATELY RECTANGULAR.

28. WS21044-XXX SELF-LOCKING NUTS ARE FOR ONE TIME USE ONLY AND MUST BE REPLACED IF REMOVED.

29. COTTER PINS ARE FOR ONE TIME USE ONLY AND MUST BE REPLACED IF REMOVED.

30. CABLE GUIDE INSTALLATION INSTRUCTIONS
NOTE: CABLE GUIDE IS TO BE INSTALLED AFTER PITCH SERVO IS INSTALLED.
1. CLAMP 115-03641-00 BRACKET TO THE CO-PILOT SIDE OF THE 115-02738-00 PITCH BRACKET APPROXIMATELY AS SHOWN IN DETAIL H.
2. SLIDE THE CO-PILOT SIDE RUDDER CABLE INTO THE CENTER SLOT OF THE 147-00326-00 GUIDE.
3. CLAMP THE 147-00326-00 GUIDE TO THE 115-03641-00 BRACKET APPROXIMATELY AS SHOWN IN VIEW T-T.
4. WITH THE RUDDER AT THE NEUTRAL POSITION, ADJUST THE GUIDE AND BRACKET LOCATIONS TO:
A. CENTER THE RUDDER CABLE WITHIN THE GUIDE AS SHOWN IN VIEW T-T.
B. LOCATE THE GUIDE ABOVE THE PITCH SERVO ELECTRICAL CONNECTOR AS SHOWN IN VIEW R-R.
C. POSITION THE GUIDE PERPENDICULAR TO THE RUDDER CABLE AS SHOWN IN VIEW R-R.
D. MAINTAIN A MINIMUM 0.3 INCH EDGE DISTANCE FOR ALL FASTENERS.
5. MATCH DRILL AND INSTALL.

31. FOR 172 S/N 80001 AND ON, AND 172S S/N 08001 AND ON, IT IS PERMISSIBLE TO TRIM THE THIRDOOD EDGE OF THE 115-02738-00 PITCH TRIM ANGLE 0.21 INCHES MAX TO ACHIEVE THE ALIGNMENT OF THE TRIM CABLE AND IDLER PULLEY AS SHOWN.

SERVO CABLE TENSION CHART

MODEL SPECIFIC INFORMATION	
EQUIPMENT	
TRIM SYSTEM RUN TIME (STOP TO STOP)	18-22 SECONDS
FLIGHT DIRECTOR COMMAND BAR GO AROUND PITCH UP ATTITUDE	7°
ESP ROLL INDICES	45° LEFT AND RIGHT
SEE INSTALLATION MANUAL, 190-02291-00 FOR POST-INSTALL CHECKOUT	
SEE ADDENDUM 005-01470-03 FOR GFC 100 CONFIGURATION SETTINGS	

EQUIPMENT LOCATION	
EQUIPMENT	ARM
GHC 507	16.6
ROLL SERVO	70.4
PITCH SERVO	175.0
PITCH TRIM SERVO	112.3

✈️ Always READ THE NOTES

✈️ Initial issuance of the 172 Model Specific Addendum only featured 29 Notes

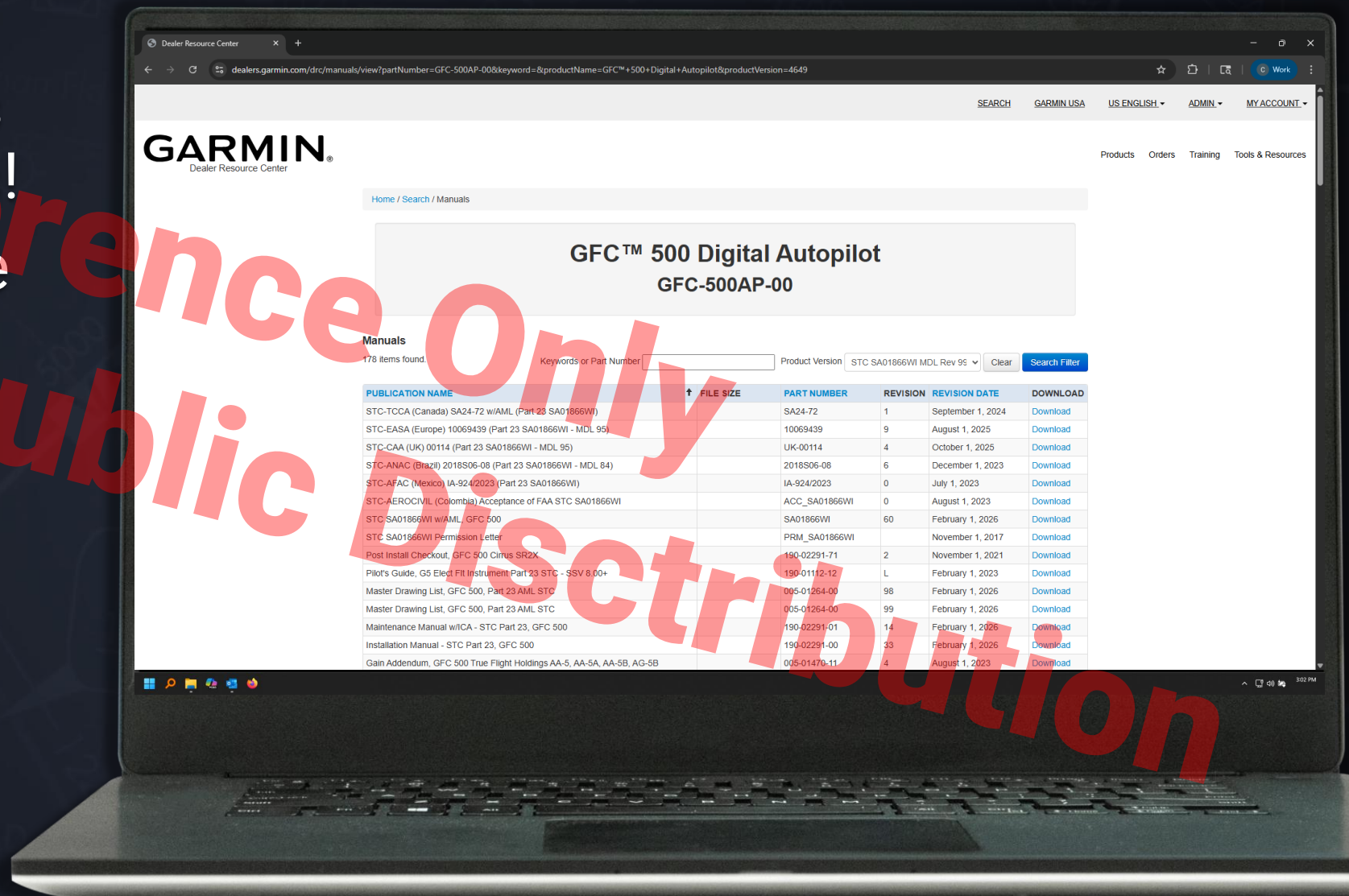
✈️ We include the lessons learned since 2017 in the notations we add to the addendum

✈️ Initial issuance also included the Gains Addendum which is now a separate document.



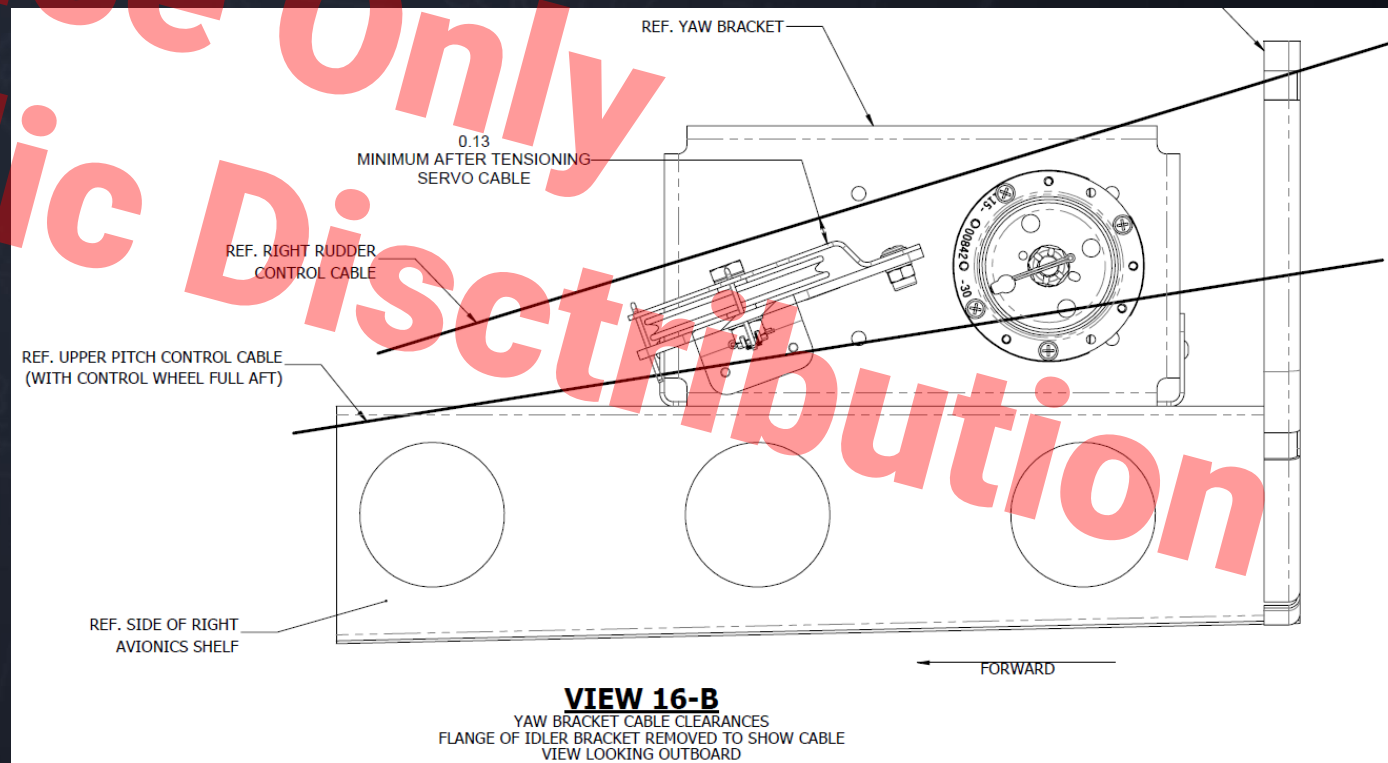
✈️ We've been talking about GFC 500 Tips and Tricks for years!

✈️ Let's revisit an oldie but a goodie....



✈ Remember this?

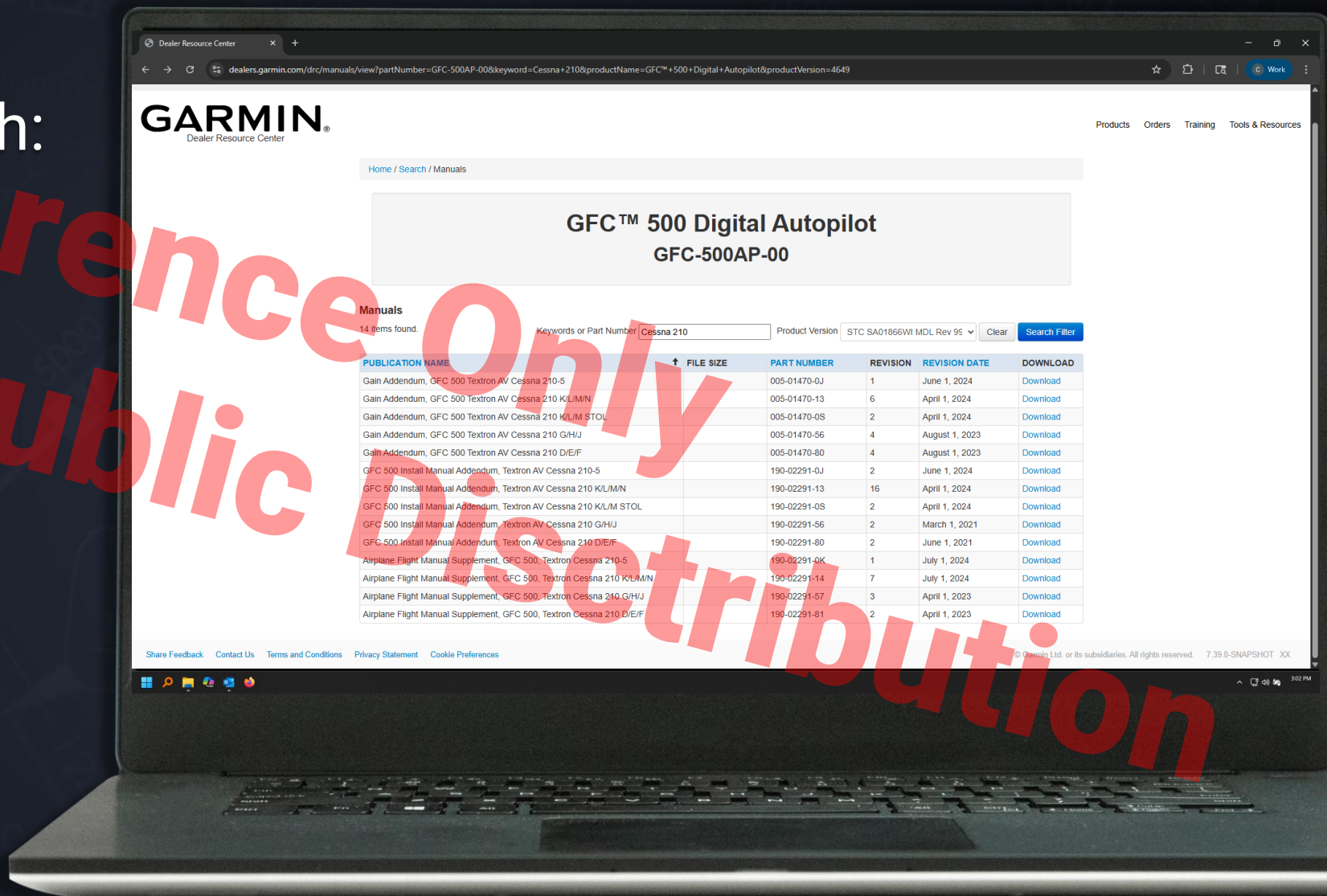
- ✈ C210 Yaw servo bracket - picture from installer. Arrows point to the elevator upper control cables.
- ✈ We use this to point out how installers sometimes don't reference the documentation correctly. There has been a revision of this information... Observe!



✈️ Sort by MDL:

✈️ Use Keyword Search:

✈️ Cessna 210



✈️ Sort by MDL:

✈️ Use Keyword Search:

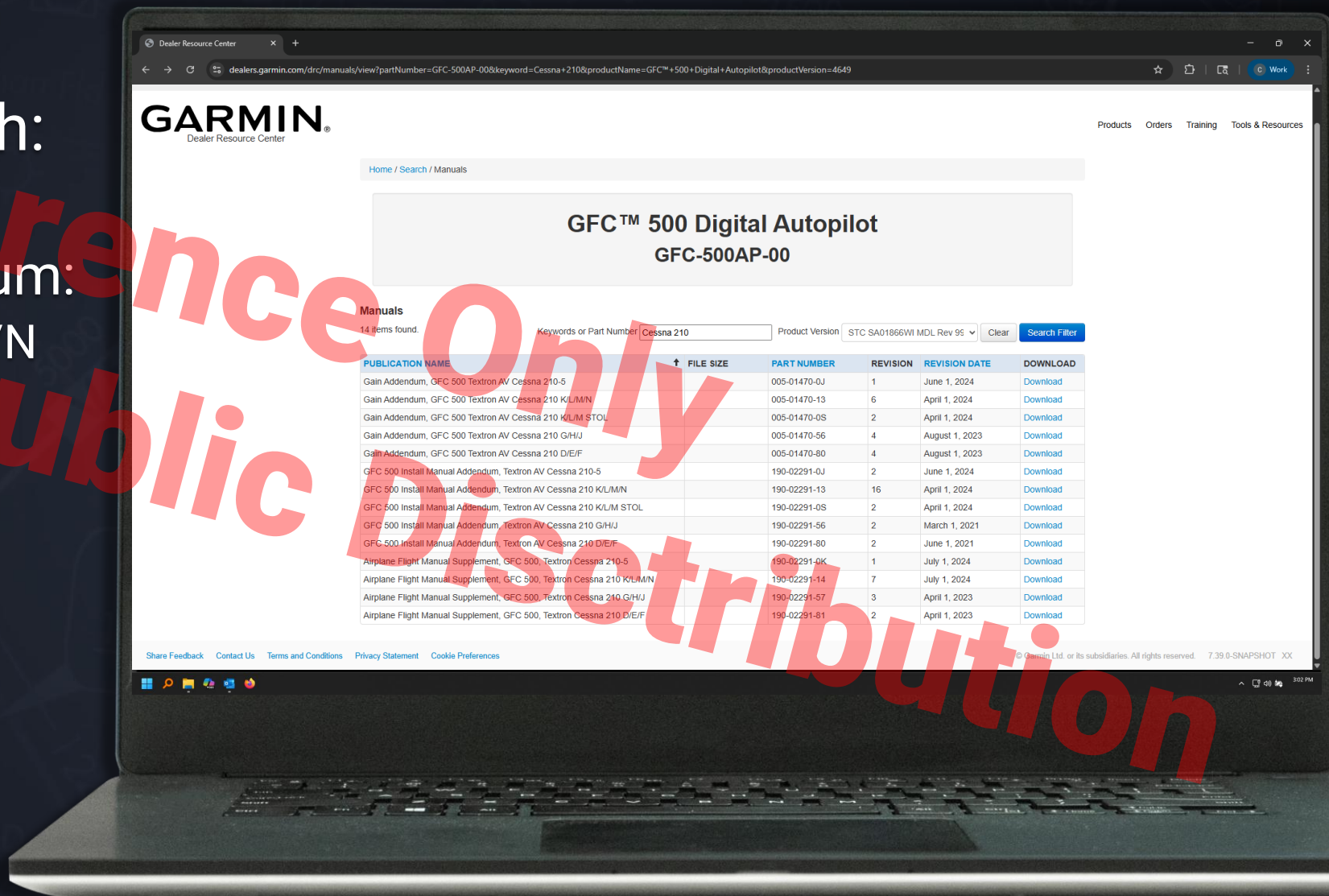
✈️ Cessna 210

✈️ Sub Model Addendum:

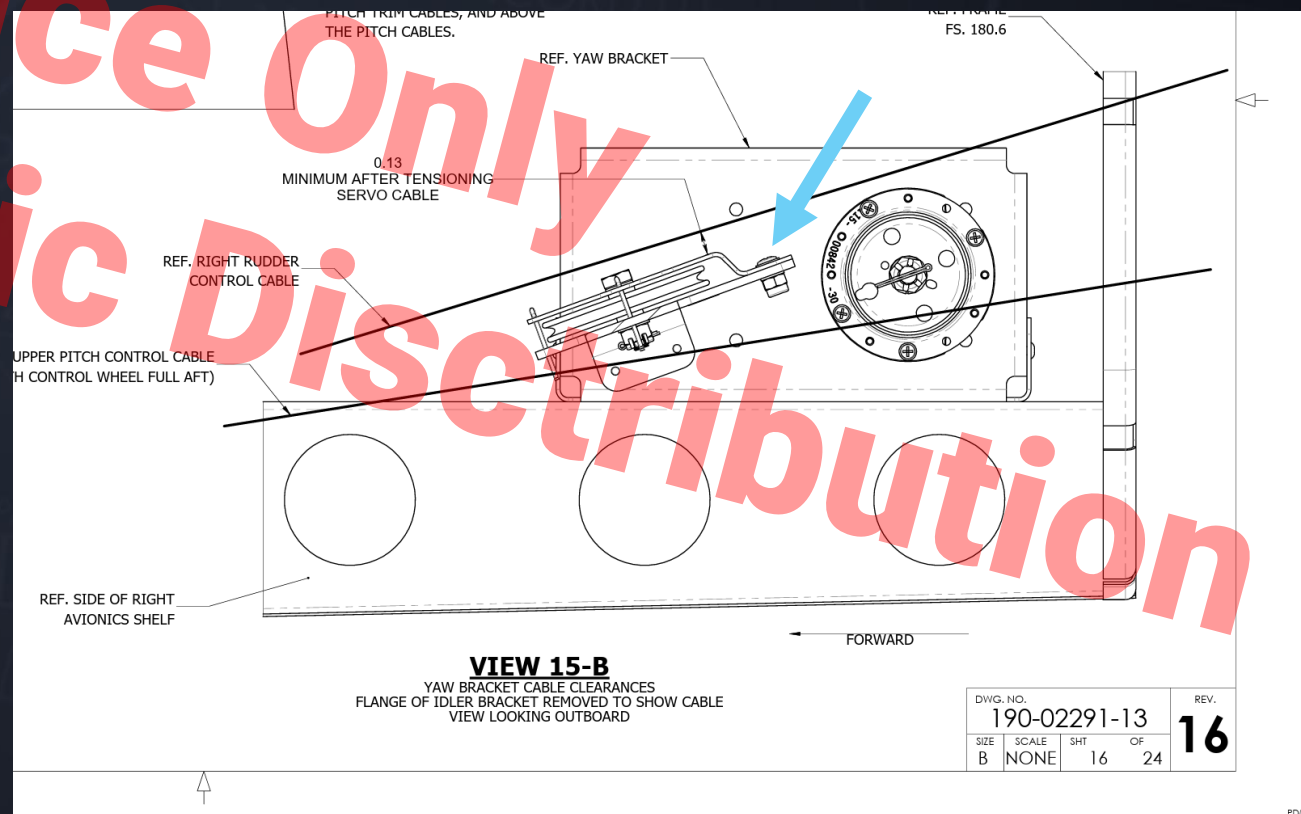
✈️ Cessna 210 K/L/M/N

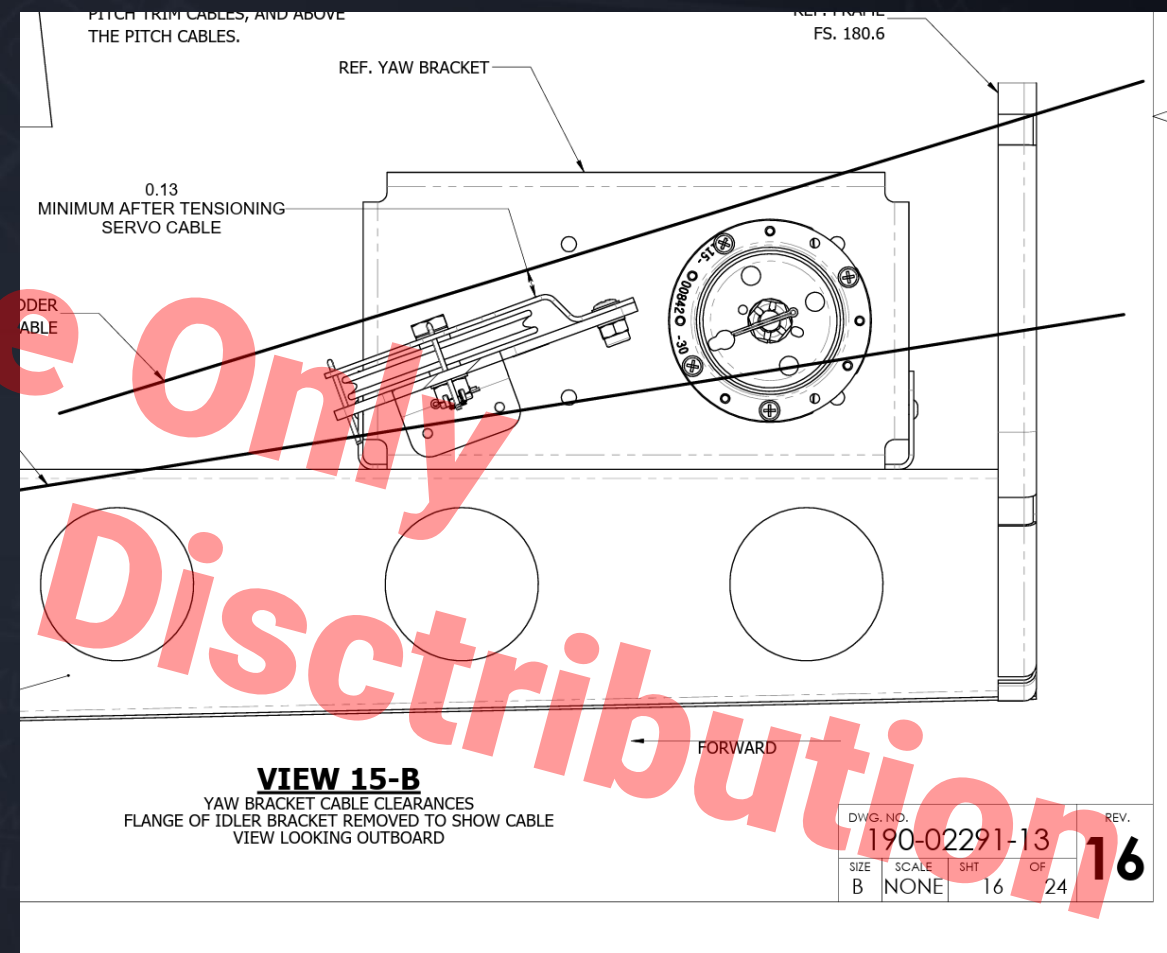
✈️ 190-02291-13

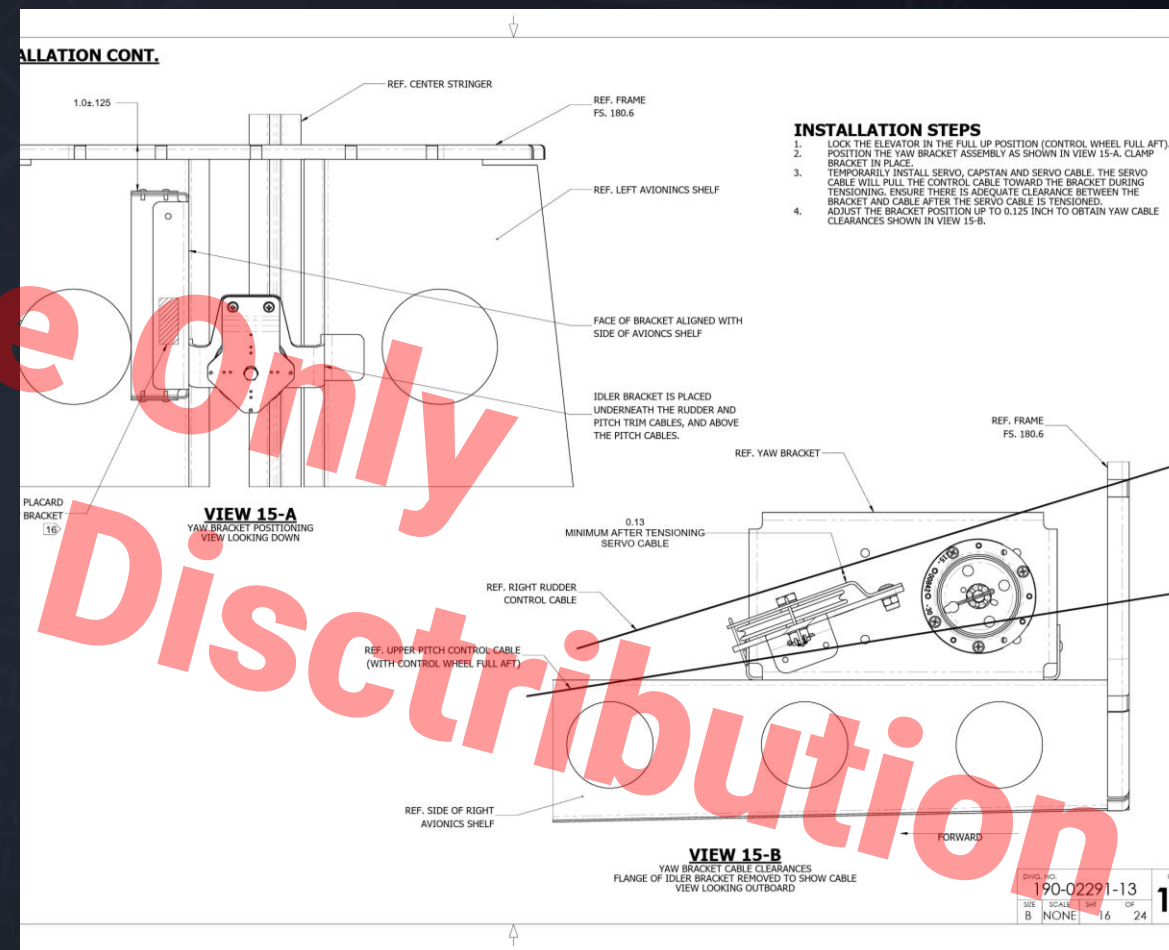
✈️ Current Rev. 16



- ✈ Latest revision: Textron Aviation Cessna 210 MSA: 190-02291-13
- ✈ Page 16: View 15-B: Still see the original orientation mistake...



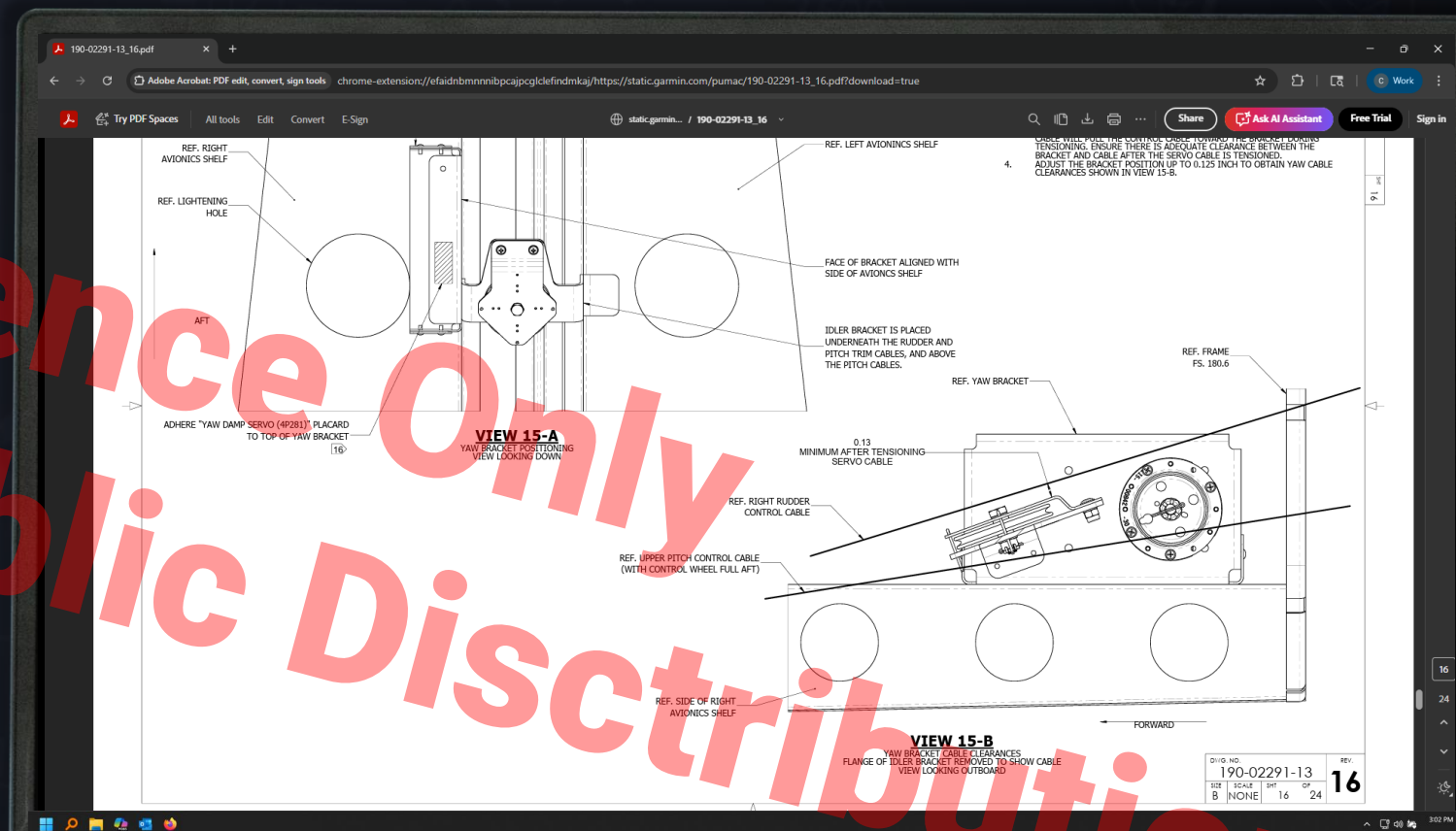




✈ Notice the newest changes to this MSA:

✈ Change of the Figures and more figures available.

✈ Let's examine...

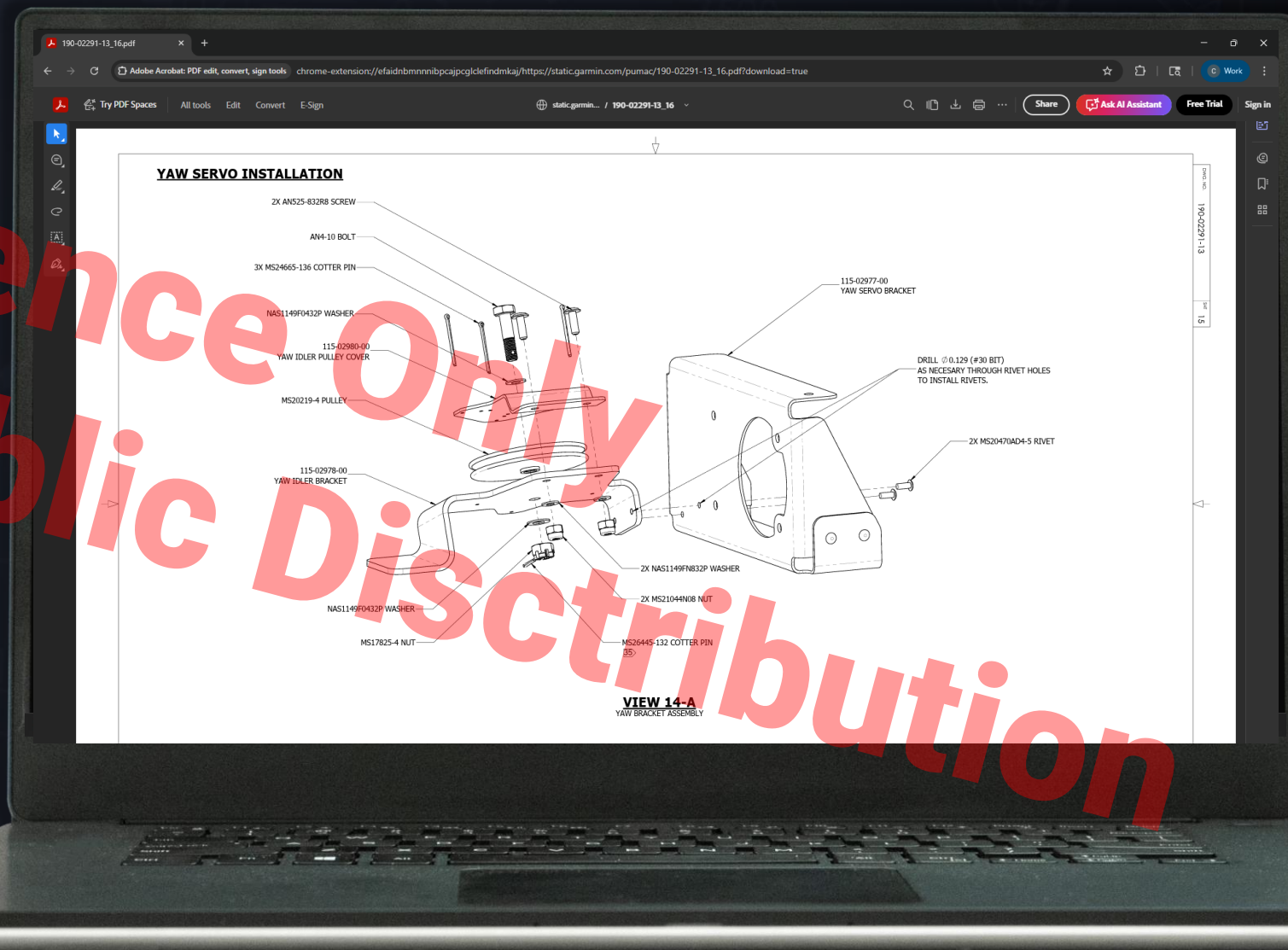


✈ Notice the newest changes to this MSA:

✈ Change of the Figure means more figures available.

✈ Let's examine...

✈ Added views to ensure installation accuracy.



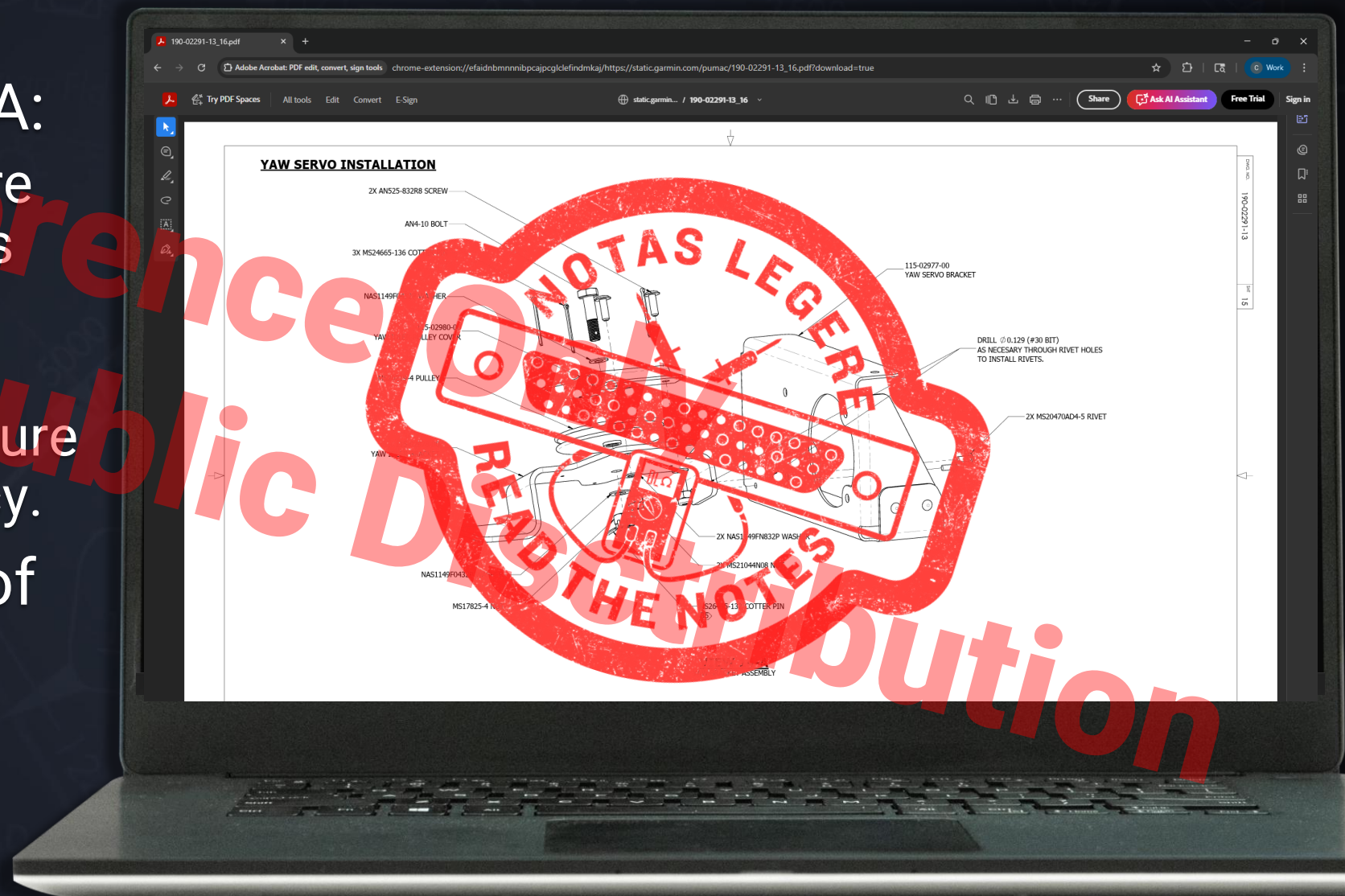
✈ Notice the newest changes to this MSA:

✈ Change of the Figure means more figures available.

✈ Let's examine...

✈ Added views to ensure installation accuracy.

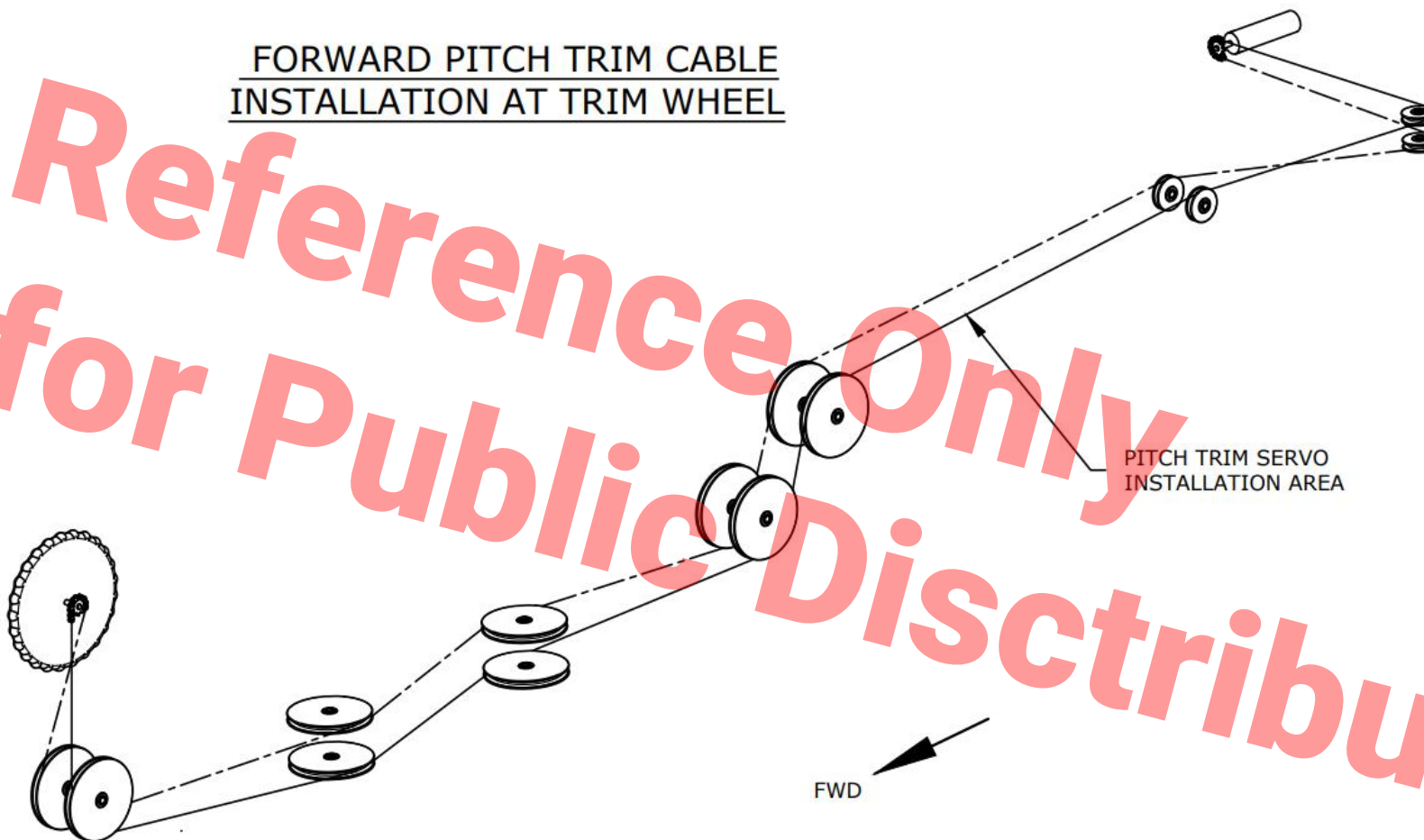
✈ Review each page of the addendum



- ✈ Pitch Trim on my Cessna 177/172/182/210/206 runs backwards?
- ✈ Software and servos are NOT the root cause.
- ✈ Line-up of the Usual Suspects:
 - ✈ Is the trim switch wiring installed correctly?
 - ✈ Are the trim servo capstan and idler are wrapped correctly?
 - ✈ Is The aircraft trim system is rigged correctly under the pedestal or in the aft fuselage.



FORWARD PITCH TRIM CABLE
INSTALLATION AT TRIM WHEEL



PITCH TRIM CABLE ROUTING

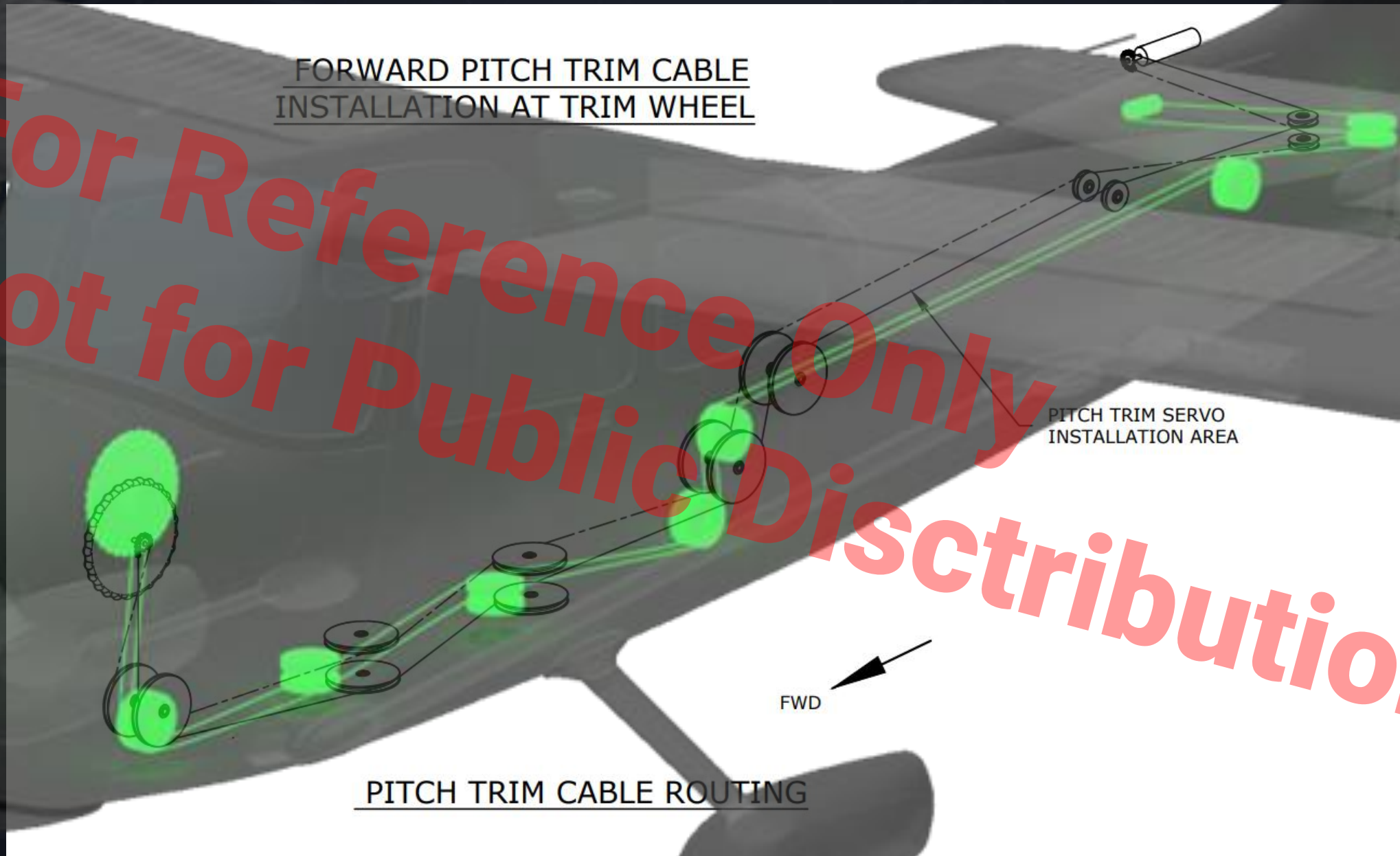
FORWARD PITCH TRIM CABLE
INSTALLATION AT TRIM WHEEL

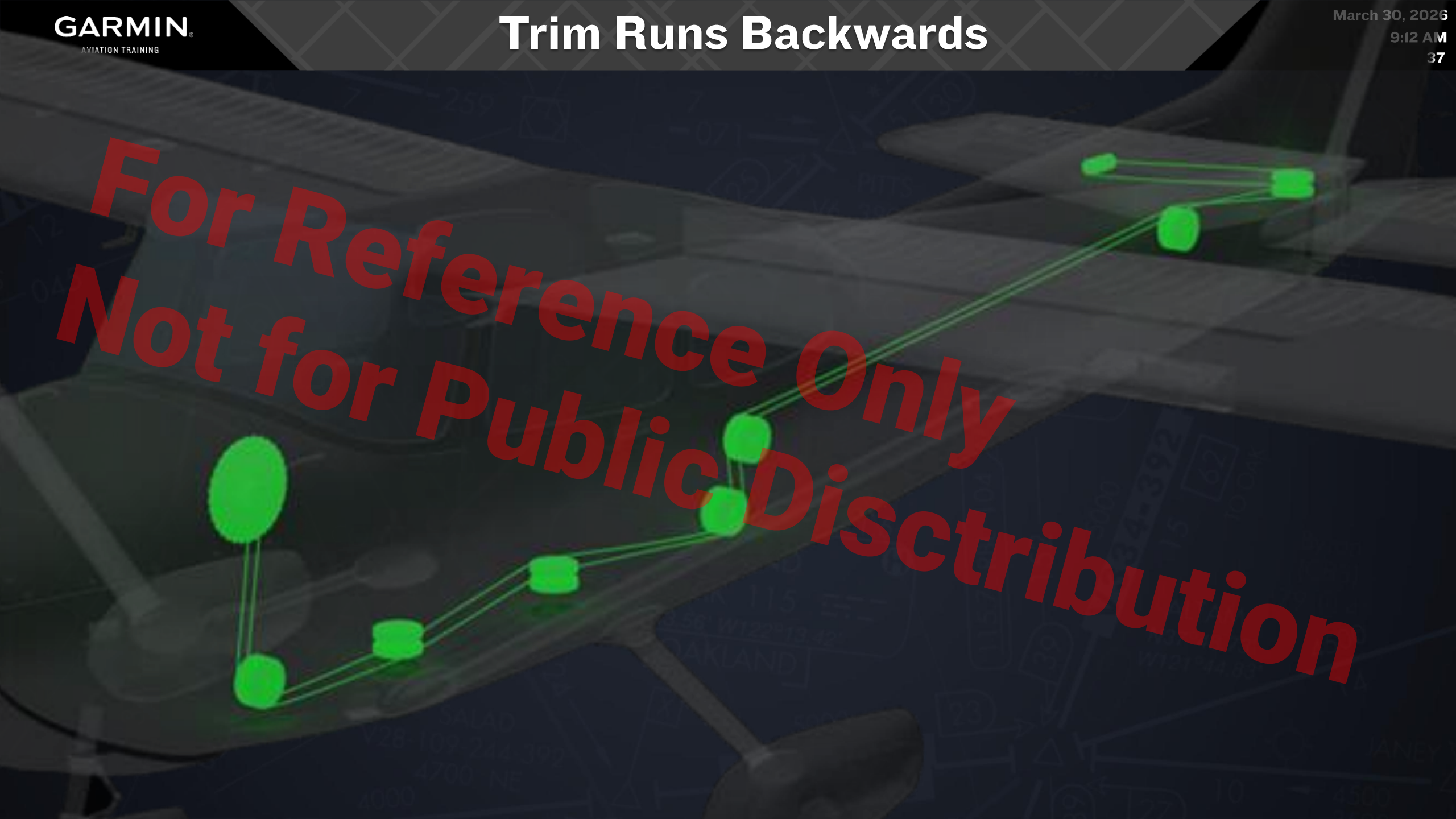
PITCH TRIM SERVO
INSTALLATION AREA

FWD

PITCH TRIM CABLE ROUTING

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Not for Public Distribution





For Reference Only
Not for Public Distribution

Trim Tab Operation

Nose Up
Trim



Trim Wheel
In Cockpit



Elevator

Trim Tab

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- ✈ Ensure your rigging offers the expected outcome

 - ✈ Servo vs Anti-servo Trim Tabs

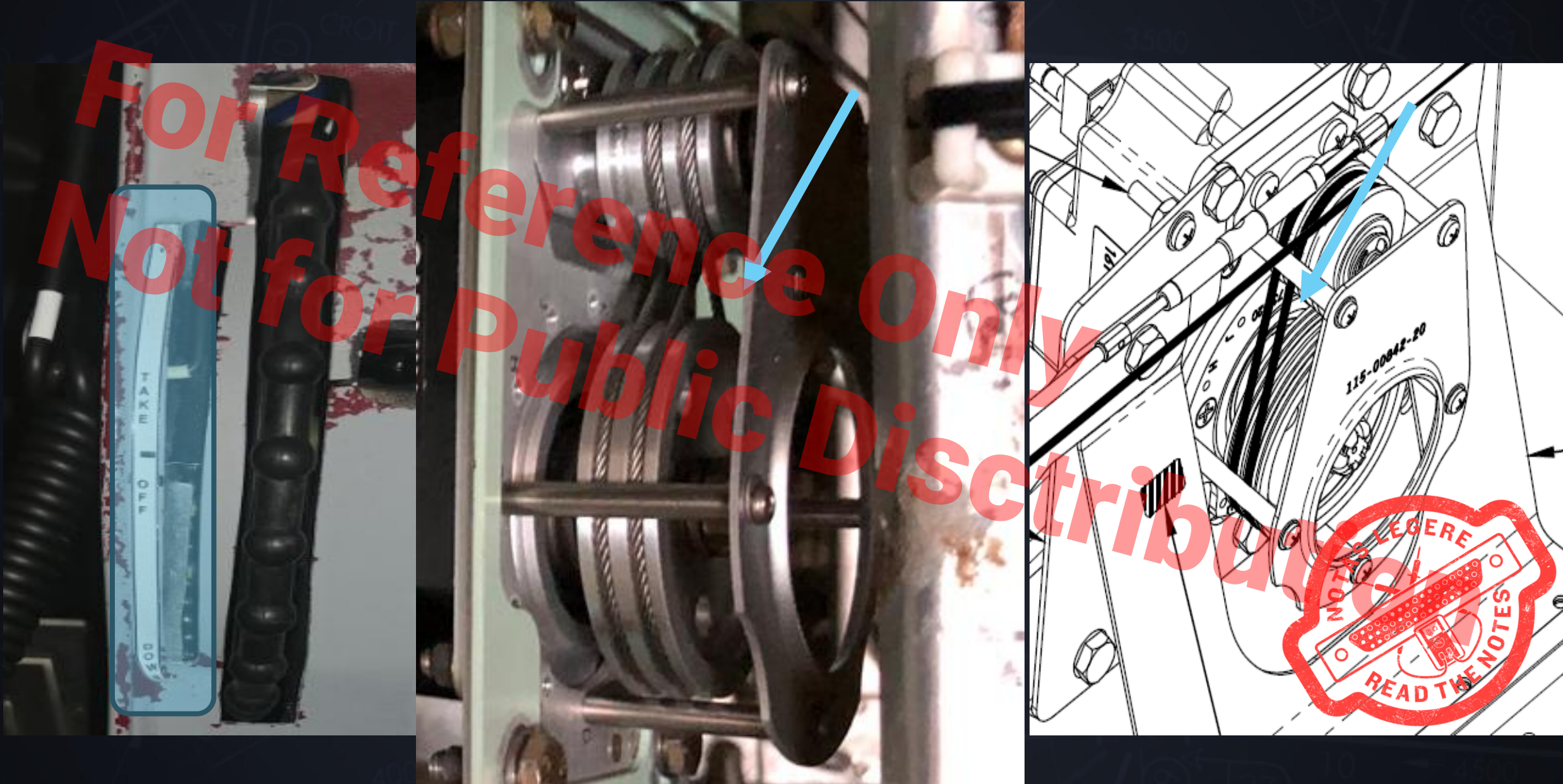
- ✈ Ensure there aren't Service Documents superseding the OEM Documentation

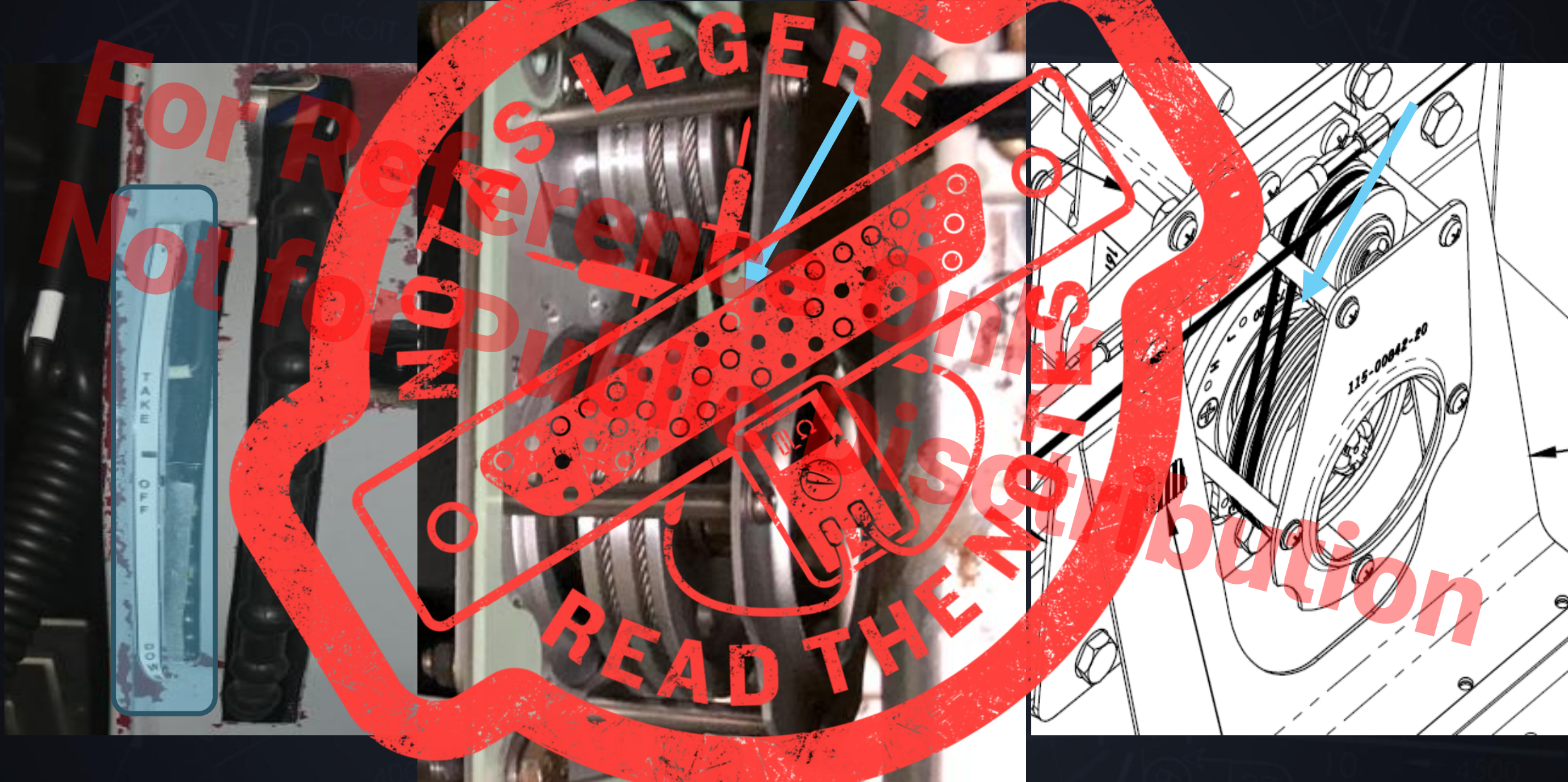
 - ✈ Cessna 195

 - ✈ Garmin Service Bulletin 24080

 - ✈ Updated Addendum (Rev 2) incorporates the SB.







✈ Every system will push software to GFC for updates

✈ Not every procedure for initial load of software matches

✈ Follow the Documentation!

✈ Read the notes!!

✈ Procedure is outlined in EFIS Installation Manuals

✈ G3X Touch

✈ G5

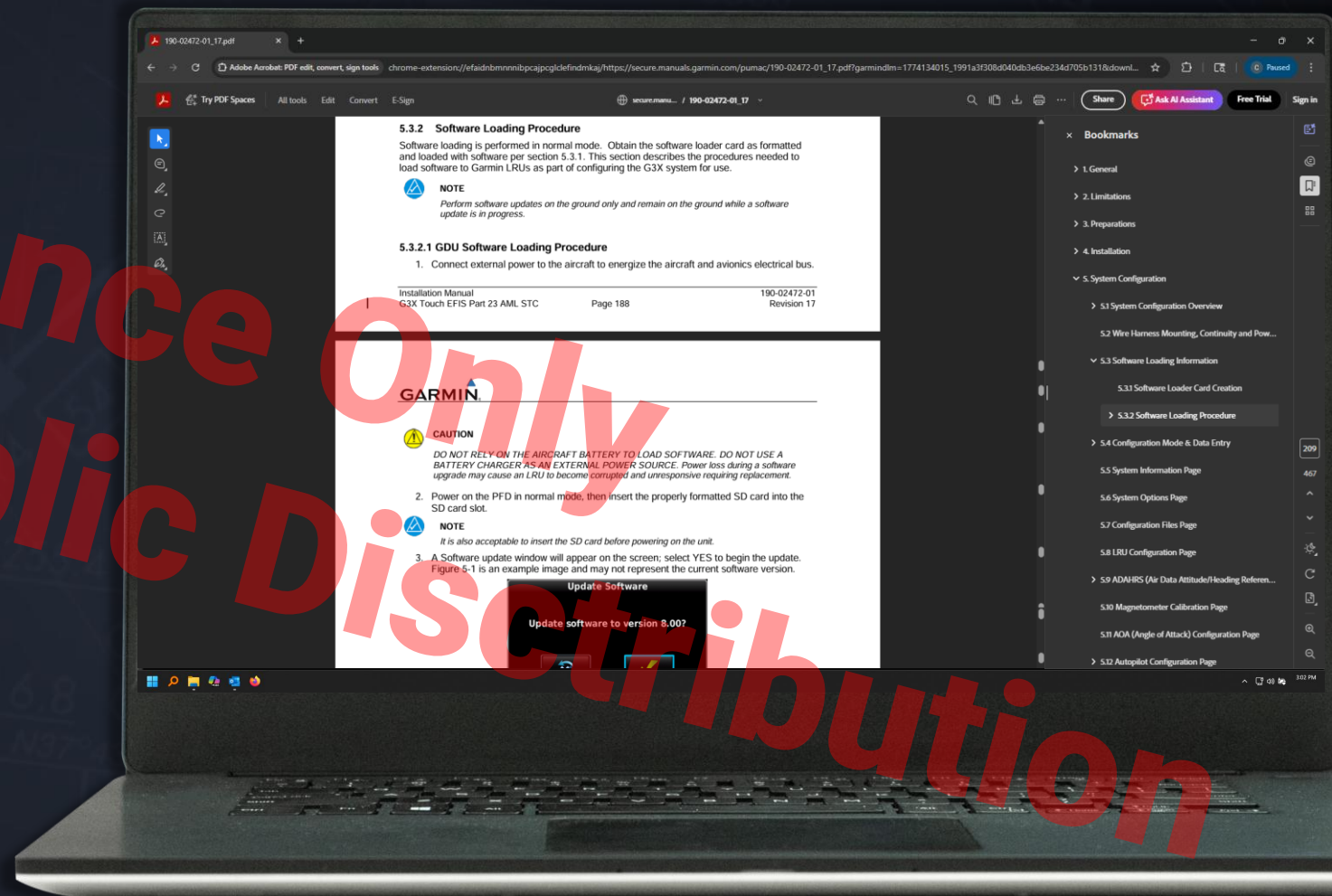
✈ GI 275



✈️ G3X Touch Software

Loading:

✈️ Section 5.3.2.1.1



✈️ G3X Touch Software Loading:

✈️ Section 5.3.2.1.1

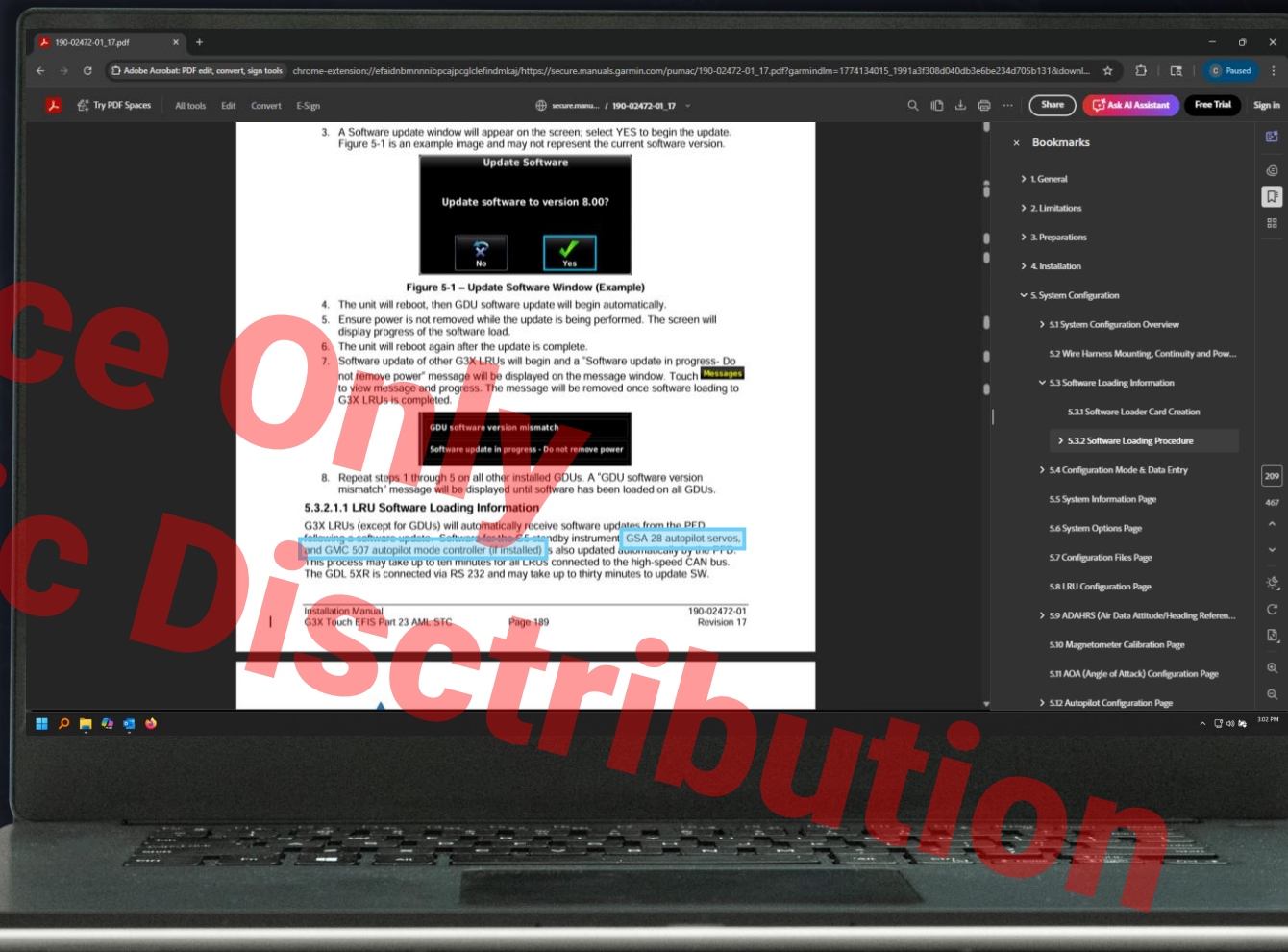
✈️ GSA 28 and GMC 507 included in EFIS Software loading and updates.

✈️ G5 works the same:

✈️ Remember: G5 was the first EFIS system for GFC 500!

✈️ GI-275:

✈️ Cannot load Software without prior configuration





NOTES

GSA 28's still ship with Software 4.30.

GI-275 will not recognize a GSA with software below 4.80.

GI-275 will not automatically push software like G3X Touch and G5 will.



- ✈ Keep in mind: GI 275 is VERY particular about sequence.
- ✈ Start with Unit ID
 - ✈ GI-1, then GI-2, and so on.
 - ✈ **RESTART REQUIRED**
- ✈ Configure GI275's Interfaced & GFC500 servos BEFORE loading software
 - ✈ Screen will have Yellow Questions Marks indicating you're ready to load software and sync the manifest



- ✈️ BEFORE loading GI 275/GFC500 Software
- ✈️ Configure all GI 275s present on each GI
 - ✈️ Interfaces > GI 275s Installed
- ✈️ Configure GFC 500 & servos present
 - ✈️ Interfaces > Autopilot > Settings
 - ✈️ Scroll to bottom, touch "Copy Config from GI 275 to GFC 500"



✈️ “GFC 500 Manifest”

- ✈️ SW/Config > Config Options > GFC500 Manifest
- ✈️ Manually sync up the reported software versions of the LRUs AFTER updating them all to the correct software level.
- ✈️ Manifest Sync does NOT load software to the GFC 500 LRUs
 - ✈️ Manifest Sync establishes the comms between the GI and GFC



- ✈ From the Software Loading Page
 - ✈ Ensure you've selected the GFC 500 packages for loading before hitting the 'Update Package' button at the bottom of the screen.



✈️ Wait for it...

✈️ Don't interrupt the software load for any given LRU in the GI-275 EFIS system

✈️ GEA

✈️ GHA

✈️ GMC

✈️ GSA

✈️ Next step?

✈️ Synchronize the Manifest!



✈ What does a Sync'd Manifest look like?

- ✈ Left column= GI275
- ✈ Right column= GFC500
- ✈ Mismatches will be highlighted yellow

✈ Notice with the mismatch:

- ✈ We're not currently communicating with the GSA's.
- ✈ Must Sync the Manifest to establish communication!



- ✈ Communication established
- ✈ Software on the GSA's match the software on the GI.



✈️ Expectation vs Reality

- ✈️ If you still see a 'Yellow Question Mark' instead of all LRU's being 'Green Check'
- ✈️ Go back to Software Loading page and reselect the GFC packages
 - ✈️ If you accomplish this task more than once and the 'Yellow Question Mark' remains?
 - ✈️ Don't forget to Manifest Sync again
 - ✈️ Still 'Yellow Question Mark'??
 - ✈️ Unplug any of the LRU's which are seen on this page as 'Green Check'



✈ Rince and repeat until all LRU's display
'Green Check'



✈ Aircraft comes in with properly functioning autopilot. I did the software upgrade. Now it's failing PFT.

✈ At what software did you start?

✈ GSA28 SLEP complied with?

✈ 'Trust but Verify' configurations

✈ Sync Manifest & Copy Config to GFC?

✈ Older installs:

✈ MOD1 GI? Mil-Spec wire?



✈ If GI 275 is having abnormal Autopilot operation or default sensors are incorrect

✈ Perform “Expansion Board Reset” a.k.a Soft Reset Procedure

✈ ‘Target’ Reload software



- ✈️ Additional Troubleshooting
 - ✈️ GI-275 Part 23 AML STC Install Manual Section 7.2





What is the greatest source of information learned during the certification process?

- A. STC Master Drawing List
- B. AML STC Installation Manual
- C. Garmin Service Bulletin or Communique
- D. Installation Manual Addendum



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After updating software, GFC now will not pass PFT.

What are my possible explanations?

- A. Did not export the configuration prior to starting update
- B. GSA28 SLEP not complied = Incorrect Part Numbers
- C. Configurations have changed upon Software update
- D. Forgot to Sync Manifest



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I have attended GFC 500 in the Red Room at AEA Nationals

- A. True
- B. False

Summary

Monday, March 30, 2026

- ✈ We've reviewed Model Specific Addendums and now understand they are the best source of information on our installs
- ✈ Reviewed several faults installers come up against and their common corrections for more accurate and efficient installs

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21A-392

GARMIN
GEA 24B
011-05291-00

412308
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Any Questions?



Thank You!

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